GEOGRAPHY

INDIAN GEOGRAPHY





INDIAN DRAINAGE SYSTEM

MODULE - 2

PENINSULAR & HIMALAYAN RIVER SYSTEM

INDIAN DRAINAGE SYSTEM

EVOLUTION OF THE HIMALAYAN DRAINAGE

POSSIBILITY OF INDO-BRAHMA RIVER

1. There are difference of opinion about the evolution of the Himalayan rivers.

2. However, geologists believe that a mighty river called Shiwalik or Indo-Brahma traversed the entire longitudinal extent of the Himalaya from Assam to Punjab and onwards to Sind.

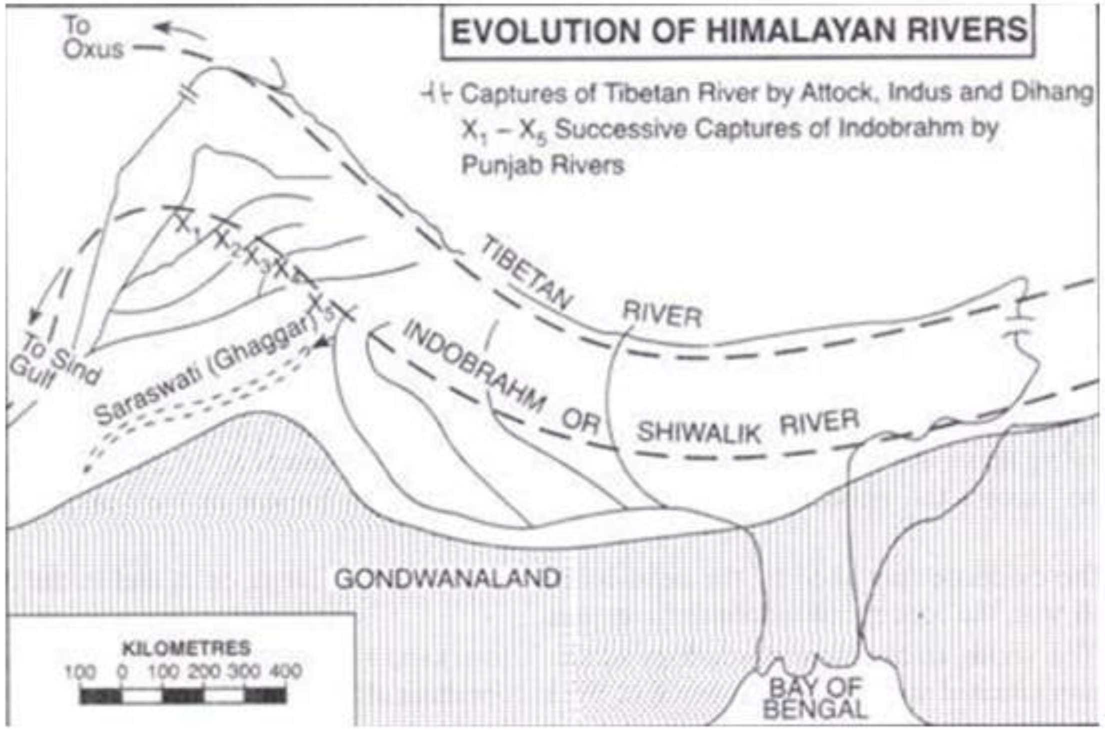
3. The remarkable continuity of the Shiwalik and its lacustrine origin and alluvial deposits support this viewpoint.

4. It is opined that in due course of time Indo– Brahma river was dismembered into three main drainage systems:
   - The Indus and its five tributaries in the western part
   - The Ganga and its Himalayan tributaries in the central part
   - The stretch of the Brahmaputra in Assam and its Himalayan tributaries in the eastern part

INDIAN DRAINAGE SYSTEM

EVOLUTION OF THE HIMALAYAN DRAINAGE

WHAT COULD HAVE CAUSED THIS DISMEMBERMENT?



5 The dismemberment was probably due to the Pleistocene upheaval in the western Himalayas, including the uplift of the Potwar Plateau along the Delhi Ridge, which acted as the water divide between the Indus and Ganga drainage systems

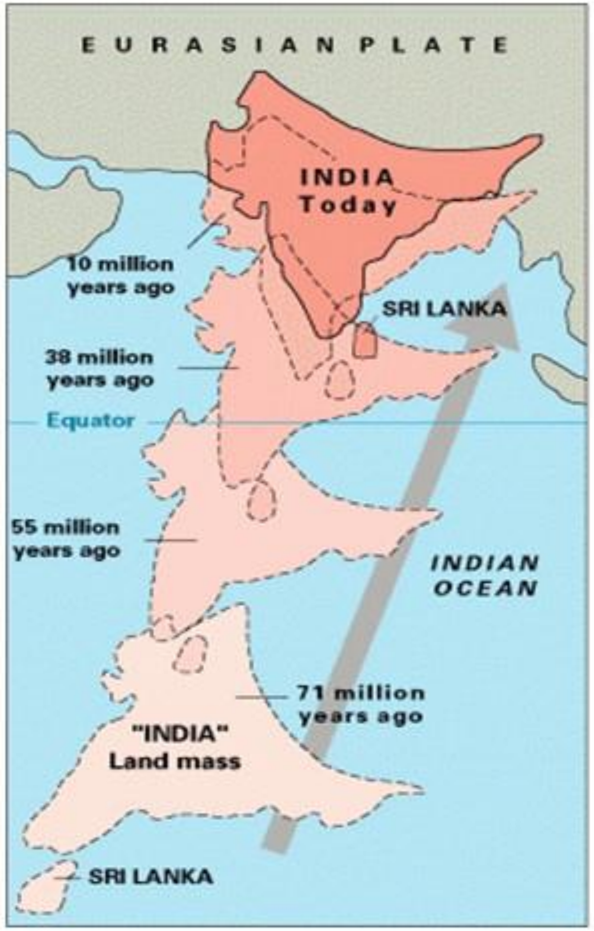
6 Likewise, the down thrusting of the Malda gap area between the Rajmahal hills and the Meghalaya plateau during the mid-Pleistocene period, diverted the Ganga and the Brahmaputra systems to flow towards the Bay of Bengal.

INDIAN DRAINAGE SYSTEM

THE PENINSULAR DRAINAGE SYSTEM

1. The Peninsular drainage system is older than the Himalayan one.
2. This is evident from the broad, largely-graded shallow valleys, and the maturity of the rivers.
3. The Western Ghats running close to the western coast act as the water divide between the major Peninsular rivers, discharging their water in the Bay of Bengal and as small rivulets joining the Arabian Sea.
4. Most of the major Peninsular rivers except Narmada and Tapi flow from west to east.
5. The Chambal, the Sind, the Betwa, the Ken, the Son, originating in the northern part of the Peninsula belong to the Ganga river system.
6. The other major river systems of the Peninsular drainage are – the Mahanadi the Godavari, the Krishna and the Kaveri.
7. Peninsular rivers are characterised by fixed course, absence of meanders and nonperennial flow of water.
8. The Narmada and the Tapi which flow through the rift valley are, however, exceptions.

THE EVOLUTION OF PENINSULAR DRAINAGE SYSTEM



Three major geological events in the distant past have shaped the present drainage systems of Peninsular India:

1. *Subsidence of Western flank of Peninsula*
2. *Upheaval of Himalayas*
3. *Slight tilting of Peninsular block from north-west to south-east direction.*

Let us now see this process in detail….

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INDIAN DRAINAGE SYSTEM

THE EVOLUTION OF PENINSULAR DRAINAGE SYSTEM

1. Subsidence of the western flank of the Peninsula leading to its submergence below the sea during the early tertiary period.

2. Generally, it has disturbed the symmetrical plan of the river on either side of the original watershed.

3. Upheaval of the Himalayas when the northern flank of the Peninsular block was subjected to subsidence and the consequent trough faulting.

4. The Narmada and the Tapi flow in trough faults and fill the original cracks with their detritus materials.

5. Hence, there is a lack of alluvial and deltaic deposits in these rivers.

6. Slight tilting of the Peninsular block from northwest to the southeastern direction gave orientation to the entire drainage system towards the Bay of Bengal during the same period.

DIFFERENCE BETWEEN THE HIMALAYAN AND THE PENINSULAR RIVER SYSTEMS

The Himalayan rivers are quite different from the rivers of the Peninsular India from the point of view of the drainage features and hydrological characteristics. A brief summary of these differences is given below:

Basis	The Himalayan River System	The Peninsular River System
1. Name	These rivers originate from the lofty Himalayan ranges and are named as the Himalayan rivers.	These rivers originate in the Peninsular Plateau and are named as Peninsular rivers.
2. Basins	These rivers have large basins and catchment areas. The total basin area of the Indus, the Ganga and the Brahmaputra is 11.78, 8.61 and 5.8 lakh sq.km respectively.	These rivers have small basins and catchment areas. The Godavari has the largest basin area of 3.12 lakh sq. km only which is less than 1/3rd the basin area of the Indus.

DIFFERENCE BETWEEN THE HIMALAYAN AND THE PENINSULAR RIVER SYSTEMS

Basis	The Himalayan River System	The Peninsular River System
3.Valleys	The Himalayan rivers flow through deep valleys called gorges. These gorges have been carved out by down cutting carried on side by side with the uplift of the Himalayas. These are examples of antecedent drainage.	The Peninsular rivers flow in comparatively shallow valleys. These are more or less completely graded valleys. The rivers have little erosional activity to perform. These are examples of consequent drainage.
4. Water Flow	The Himalayan rivers are perennial in nature, i.e., the water flows throughout the year in these rivers. These rivers receive water both from the monsoons and snow melt. The perennial nature of these rivers make them useful for irrigation.	The Peninsular rivers receive water only from rainfall and water flows in these rivers in rainy season only. Therefore, these rivers are seasonal or non-perennial. As such these rivers are much less useful for irrigation.
5. Stage	These rivers flow across the young fold mountains and are still in a youthful stage.	These rivers have been flowing in one of the oldest plateaus of the world and have reached maturity.

DIFFERENCE BETWEEN THE HIMALAYAN AND THE PENINSULAR RIVER SYSTEMS

Basis	The Himalayan River System	The Peninsular River System
6. Meanders	The upper reaches of the Himalayan rivers are highly tortuous. When they enter the plains, there is a sudden reduction in the speed of flow of water. Under these circumstances these rivers form meanders and often shift their beds.	The hard rock surface and non-alluvial character of the plateaus permits little scope for the formation of meanders. As such, the rivers of the Peninsular Plateau follow more or less straight courses.
7. Deltas and Estuaries	The Himalayan rivers form big deltas at their mouths. The Ganga-Brahmaputra delta is the largest in the world.	Some of the Peninsular rivers, such as the Narmada and the Tapi form estuaries. Other rivers such as the Mahanadi, the Godavari, the Krishna, and the Cauvery from deltas . Several small streams originating from the Western Ghats and flowing towards the west enter the Arabian Sea without forming any delta.

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

The Himalayan drainage consists of several river systems but the following are the major river systems:

1 THE INDUS SYSTEM

1. It is one of the largest river basins of the world, with a total length of 2,880 km (in India 1,114 km).
2. The Indus also known as the Sindhu, is the westernmost of the Himalayan rivers in India.
3. It originates from a glacier near Bokhar Chu (31°15' N latitude and 81°40' E longitude) in the Tibetan region at an altitude of 4,164 m in the Kailash Mountain range.

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

The Himalayan drainage consists of several river systems but the following are the major river systems:

1 THE INDUS SYSTEM

AREA → Covering an area of 11,65,000 sq. km (in India it is 321, 289 sq. km and a total length of 2,880 km.

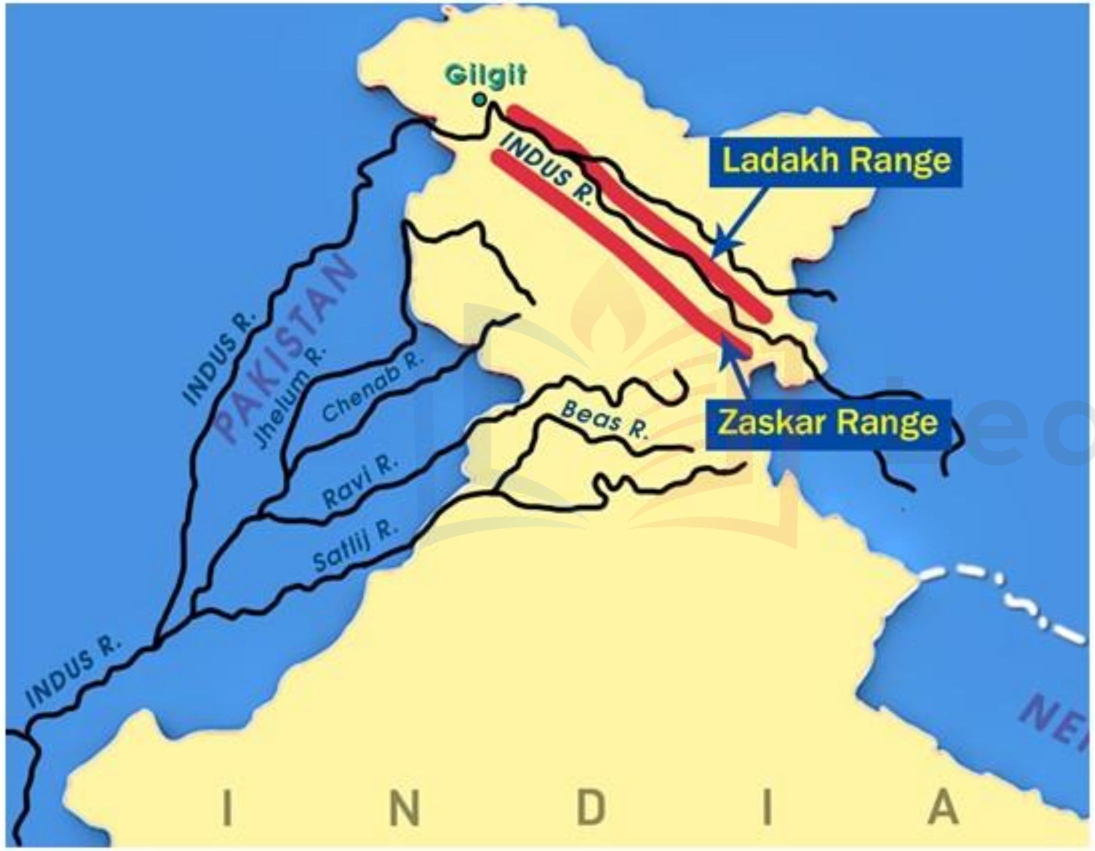
NAME OF THE RIVER → Also known as the Sindhu. In Tibet, it is known as 'Singi Khamban; or Lion's mouth.

ORIGIN OF THE RIVER → A glacier near Bokhar Chu in the Tibetan region in the Kailash Mountain range (near Manasarovar lake).

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

1 THE INDUS SYSTEM



FLOW OF THE RIVER

1. After flowing in the northwest direction between the Ladakh and Zaskar ranges, it passes through Ladakh and Baltistan.
2. It cuts across the Ladakh range, forming a spectacular gorge near Gilgit in Jammu and Kashmir.
3. It enters into Pakistan near Chillar in the Dardistan region.

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

1 THE INDUS SYSTEM

Indus gorge in Gilgit

Dardistan region

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

1 THE INDUS SYSTEM

- The Indus receives a number of tributaries such as the Shyok, the Gilgit, the Zaskar, the Hunza, the Shigar, the Gasting and the Dras.
- It also receives some other tributaries like the Khurram, the Tochi, the Gomal, the Viboa and the Sangar.
- They all originate in the Sulaiman ranges.

IMPORTANT RIGHT BANK TRIBUTARIES

Shyok
Gilgit
Hunza river
Khurram river
Gomal river etc...

Not important for exams

IMPORTANT LEFT BANK TRIBUTARIES

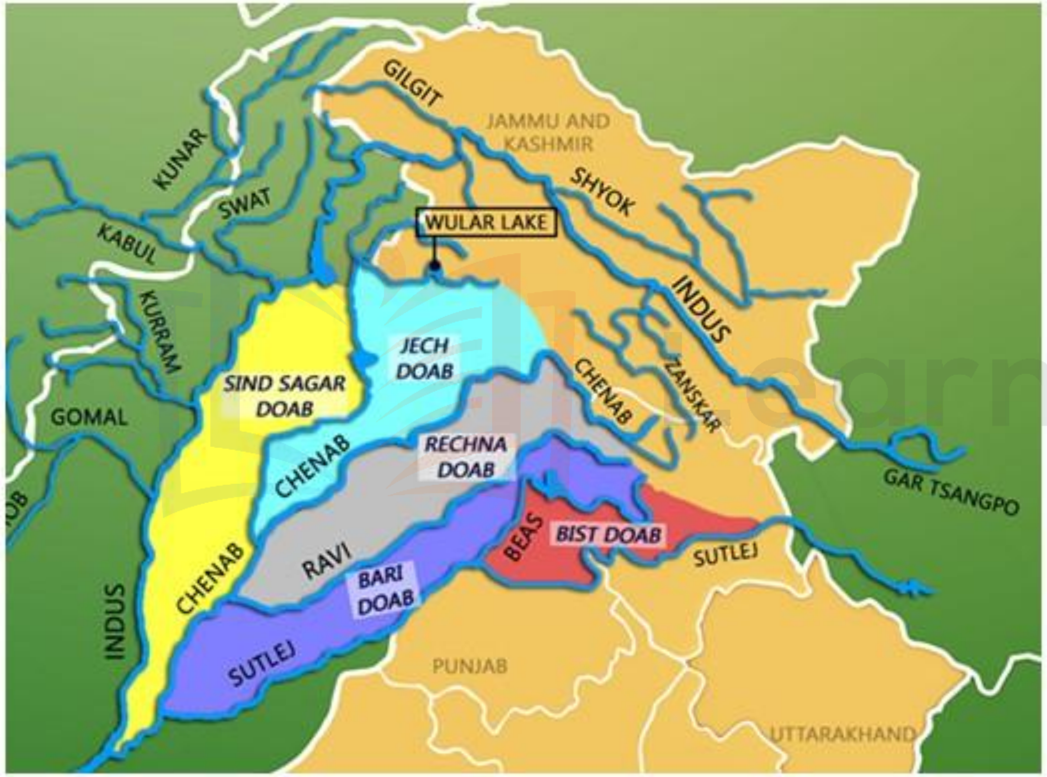
Zanskar
The Panjnad rivers

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

1 THE INDUS SYSTEM

THE PANJNAD RIVERS



1. The river flows southward and receives 'Panjnad' a little above Mithankot.
2. The Panjnad is the name given to the five rivers of Punjab, namely the Satluj, the Beas, the Ravi, the Chenab and the Jhelum.
3. It finally discharges into the Arabian Sea, east of Karachi.
4. The Indus flows in India only through the Leh district in Jammu and Kashmir.

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

1 THE INDUS SYSTEM

JHELUM

5 The Jhelum, an important tributary of the Indus, rises from a spring at Verinag situated at the foot of the Pir Panjal in the south-eastern part of the valley of Kashmir.

6 It flows through Srinagar and the Wular lake before entering Pakistan through a deep narrow gorge

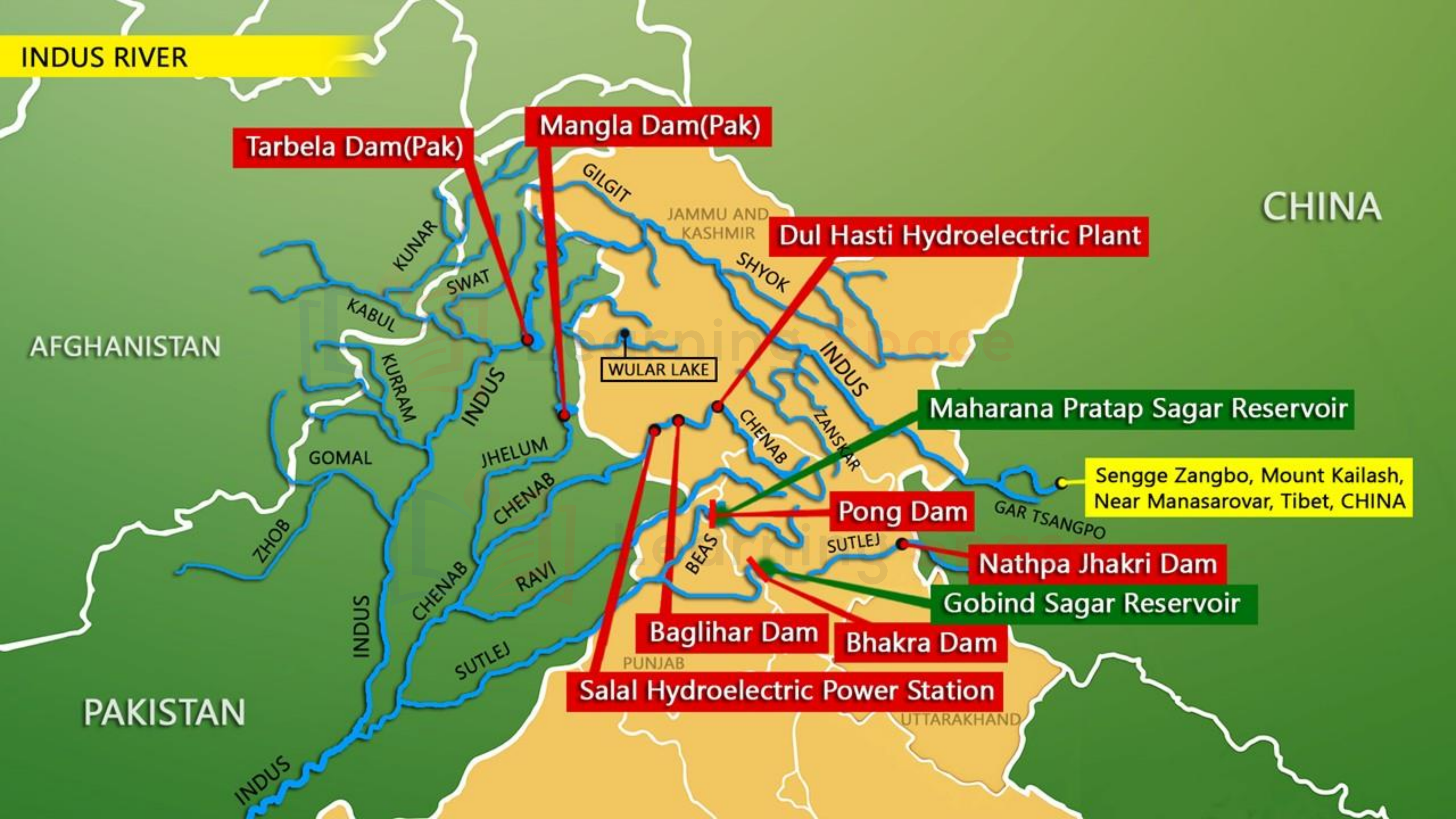
7 It joins the Chenab near Jhang in Pakistan.

CHENAB

8 The Chenab is the largest tributary of the Indus.

9 It is formed by two streams, the Chandra and the Bhaga, which join at Tandi near Keylong in Himachal Pradesh.

10 Hence, it is also known as Chandrabhaga. The river flows for 1,180 km before entering into Pakistan.

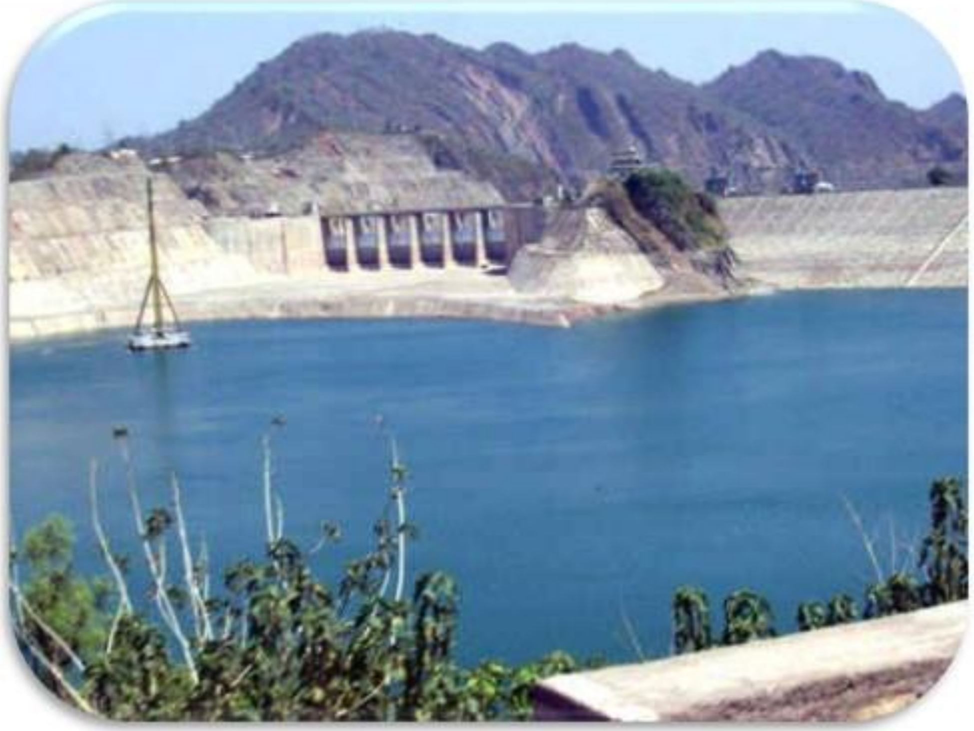

INDUS RIVER
Tarbela Dam(Pak)
Mangla Dam(Pak)
GILGIT
JAMMU AND KASHMIR
Dul Hasti Hydroelectric Plant
CHINA
KUNAR
SWAT
KABUL
SHYOK
AFGHANISTAN
KURRAM
INDUS
WULAR LAKE
INDUS
Maharana Pratap Sagar Reservoir
CHENAB
ZANSKAR
GOMAL
JHELUM
Sengge Zangbo, Mount Kailash, Near Manasarovar, Tibet, CHINA
CHENAB
GAR TSANGPO
Pong Dam
ZHOB
BEAS
SUTLEJ
Nathpa Jhakri Dam
RAVI
CHENAB
Gobind Sagar Reservoir
INDUS
Baglihar Dam
Bhakra Dam
SUTLEJ
PUNJAB
Salal Hydroelectric Power Station
PAKISTAN
UTTARAKHAND
INDUS

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

1 THE INDUS SYSTEM

RAVI



Thein dam aka Ranjit Sagar dam is present across river Ravi.

11 The Ravi is another important tributary of the Indus.

12 It rises west of the Rohtang pass in the Kullu hills of Himachal Pradesh and flows through the Chamba valley of the state.

13 Before entering Pakistan and joining the Chenab near Sarai Sidhu, it drains the area lying between the southeastern part of the Pir Panjal and the Dhauladhar ranges.

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

1 THE INDUS SYSTEM

BEAS

- **Pong Dam is situated across Beas River.**
- **Harike barrage – meeting point of Beas and Sutlaj rivers**

14 The Beas is another important tributary of the Indus, originating from the Beas Kund near the Rohtang Pass at an elevation of 4,000 m above the mean sea level.

15 The river flows through the Kullu valley and forms gorges at Kati and Largi in the Dhaoladhar range.

16 It enters the Punjab plains where it meets the Satluj near Harike.

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INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

1 THE INDUS SYSTEM

SATLUJ

27 The Satluj originates in the Rakas lake near Mansarovar at an altitude of 4,555 m in Tibet where it is known as Langchen Khambab.

28 It flows almost parallel to the Indus for about 400 km before entering India, and comes out of a gorge at Rupar.

29 It passes through the Shipki La on the Himalayan ranges and enters the Punjab plains. It is an antecedent river.

30 It is a very important tributary as it feeds the canal system of the Bhakra Nangal project.

INDIAN DRAINAGE SYSTEM

THE INDUS DRAINAGE SYSTEM

Name of the river	Source	Length in km	Area drained (sq km)
Indus	Near Manasarovar Lake	2,880 (709 in India)	1,178,440 (321,290 in India)
Jhelum	Verinag at 4,900 m altitude	724	34,775 upto Indo-Pak border
Chenab	Bara Lacha Pass	1,180	26,155 upto Indo-Pak Border
Ravi	Near Rohtang Pass	725	14,442 (5,957 in India)
Beas	Near Rohtang Pass 4,062 m height	460	20,303
Satluj	Manasarovar-Rakas Lakes at 4,570 m altitude	1,450 (1,050 in India)	25,900

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